

Unit 6 - Operations with Integers

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The following B.E.S.T Standards will be covered in this unit:

MA.6.NSO.3.3 Evaluate positive rational numbers and integers with natural number exponents.

- *Clarification 1: Within this benchmark, expectations include using natural number exponents up to 5.*

MA.6.NSO.3.4 Express composite whole numbers as a product of prime factors with natural number exponents.

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.

- *Clarification 1: Instruction begins with the use of manipulatives, models and number lines working towards becoming procedurally fluent by the end of grade 6.*
- *Clarification 2: Instruction focuses on the inverse relationship between the operations of addition and subtraction. If p and q are integers, then $p - q = p + (-q)$ and $p + q = p - (-q)$.*

MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.

- *Clarification 1: Instruction includes the use of models and number lines and the inverse relationship between multiplication and division, working toward becoming procedurally fluent by the end of grade 6.*
- *Clarification 2: Instruction focuses on the understanding that integers can be divided, provided that the divisor is not zero, and every quotient of integers (with non-zero divisor) is a rational number. If p and q are integers where $q \neq 0$, then $-\left(\frac{p}{q}\right) = \frac{-p}{q} = \frac{p}{-q}$ and $\frac{p}{q} = \frac{-p}{-q}$.*

Unit 6, Lesson 1: Exponents



Benchmark

MA.6.NSO.3.3 Evaluate positive rational numbers and integers with natural number exponents.



Learning Targets

- ☐ I understand the role of a base and an exponent in the exponential form of a numeric expression.
- ☐ I can write a numeric expression using repeated multiplication in exponential form.
- ☐ I can write an expression in exponential form as a numeric expression using repeated multiplication.
- ☐ I can evaluate a positive rational number raised to a power.



6.1.1 Warm-Up: Graphic Designer

1. Maurice Woods is a graphic designer and founder of the Inneract Project. He says, "Design is really about a process of thinking creatively and critically about how to solve something in a way that not only makes sense to people, but produces a solution to a problem." How does that process relate to your study of mathematics?
2. Maurice states that one of the skills he brought with him from his basketball career to his design career was his experience with teamwork and what it means to be flexible and compromise. Describe an experience when teamwork was needed for you to reach a goal.
3. Maurice said, "I want to be remembered for giving back to the community and giving back to young people ... I get more joy out of that than any project I work on." What do you want to be remembered for in life?





6.1.2 Guided Instruction: Expressions with Exponents

1. The principal at Fitzgerald Middle School has been tasked with contacting everyone at the school regarding an upcoming change. The principal uses a phone tree to contact the staff. The principal calls 2 people, then those 2 people will call 2 other people. This process repeats until everyone has been contacted.

- a. Fill in the blanks for the phone tree.

Phone Tree	
First round of calls:	2
Second round of calls:	$2 \cdot 2$
Third round of calls:	$2 \cdot 2 \cdot 2$
Fourth round of calls:	$2 \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}}$
Fifth round of calls:	

- b. What do you notice about the phone tree?
- c. How many people were contacted on the fifth round?

The fifth round of calls can be rewritten in exponential form as 2^5 .

- d. How does the exponential expression for the fifth round of calls relate to the repeated multiplication written in the phone tree?

The numeric expression $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$ can be rewritten in exponential form as 2^5 , where 2 is the **base** and 5 is the **exponent**.



The **base** of an exponent is the number used as a factor in exponential form.



The **exponent** is the number of times the base occurs as a factor.

Exponents are also commonly referred to as powers. For example, 2^5 is read as "2 to the 5th power."

2. Complete the statements using the definitions of base and exponent.

- a. In the exponential expression 9^3 , the 9 is the

- ☐ base
- ☐ exponent

of the expression and the 3 is the

- ☐ base.
- ☐ exponent.

- b. The exponential expression 9^3 has the same value

as

- ☐ $3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3$.
- ☐ $9(3)$.
- ☐ $(9)(9)(9)$.



In mathematics, **evaluate** is an instruction that means "find the value of."

3. Evaluate 9^3 .



6.1.3 Guided Practice: Equivalent Expressions

1. Write each numeric expression in exponential form.

a. $5 \cdot 5 \cdot 5 \cdot 5$

b. $\left(\frac{6}{7}\right) \times \left(\frac{6}{7}\right) \times \left(\frac{6}{7}\right)$

c. $(3.25)(3.25)$

2. Rewrite each expression given in exponential form as repeated multiplication.

a. 7^5

b. $(1.5)^2$

c. $\left(\frac{3}{4}\right)^3$



6.1.4 Collaboration: Evaluating Exponential Expressions

1. With your partner, discuss the four expressions in the table.

Expression A	$(2)(2)(2)(2)$
Expression B	(2^4)
Expression C	16
Expression D	2×4

a. Which expression does not represent the same value as the others?

b. Explain your answer or show the work that led to your conclusion.

c. Explain what mathematics process should be done if the directions state to **evaluate** an expression.



2. Marta's work for evaluating the expression 3^2 is shown:

$$3^2 = 3 \times 2 = 6$$

- Discuss Marta's work with your partner.
- Explain whether her work is correct or incorrect. If it is correct, describe how you know. If it is incorrect, describe her error and provide the correct evaluation of the expression.

3. Determine if each equation in the table is true or false. Show your work to support your answer.

Equation	True or False?	Work
$16^1 = 8^2$		
$3 + 3 + 3 = 3^3$		
$\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = 4 \cdot \frac{1}{2}$		
$(5)^3 = 5 \cdot 5 \cdot 5$		



6.1.5 Try It: Evaluating Exponential Expressions

1. Evaluate each expression written in exponential form.

a. $\left(\frac{2}{5}\right)^2$

b. 4^5

c. $(6.2)^3$



6.1.6 Your Turn!

1. For each set of expressions, circle the equivalent expressions.

a. $(4)^3$ $(3)^4$ $(4)(4)(4)$ $4 + 4 + 4$

b. $\left(\frac{5}{3}\right)^2$ $\left(\frac{3}{5}\right)^2$ $\frac{10}{6}$ $\frac{25}{9}$

2. Complete the table by rewriting the numeric expression in the form indicated. Then, find the value of each expression.

Exponential Form	Expanded Form	Value
	$(11)(11)(11)$	
$\left(\frac{7}{15}\right)^3$		
$(5.2)^2$		
	$\frac{4}{5} \cdot \frac{4}{5} \cdot \frac{4}{5} \cdot \frac{4}{5}$	



6.1.7 Wrap-Up: Ticket Out the Door

Jose and Georgette were asked to each roll a number cube five times and record the number from each roll onto a chart. The number that Jose rolls represents the base of an exponential expression, and Georgette's roll represents the exponent of the expression. The table below shows the results of rolling the number cubes five times.

Roll	Jose's Number Cube	Georgette's Number Cube	Exponential Expression
1	5	3	5^3
2	5	4	5^4
3	6	2	6^2
4	4	5	4^5
5	1	4	1^4

Jose conjectures that the value of the exponential expression from Roll 3 is the greatest because the exponential expression that has the largest base is always greater.

Georgette conjectures that the value of the exponential expression from Roll 4 is the greatest because the exponential expression that has the largest exponent is always greater.

- Determine if either Jose or Georgette were right. If neither are correct, then which exponential expression has the largest value?
- Determine if it is possible that Jose's conjecture or Georgette's conjecture is always true using the exponential expressions 2^5 and 5^2 . Explain your findings.

Unit 6, Lesson 2: Prime Factorization



Benchmark

MA.6.NSO.3.4 Express composite whole numbers as a product of prime factors with natural number exponents.



Learning Target

- ☐ I can express composite whole numbers as a product of prime factors.



6.2.1 Warm-Up: Would You Rather?

- Which of the following would you rather choose? Justify your reasoning using mathematics.

Deposit \$3 in the bank and have it triple each week for 4 weeks.

Deposit \$4 in the bank and have it quadruple each week for 3 weeks.



6.2.2 Collaboration: Exploring Factors

Work with your partner on the following.

1. Write the number 24 as a factor pair that uses the greatest factor other than itself.
2. Write the number 12 as a factor pair that uses the greatest factor other than itself.
3. Write the number 6 as a factor pair that uses the greatest factor other than itself.
4. Write the number 3 as a factor pair that uses the greatest factor other than itself.
5. How did each of the above tasks, problems 2-4, relate to the factor pair that came before it?
6. What was different about rewriting the number 3 as a factor pair using the greatest factor other than itself?



6.2.3 Guided Instruction: Prime Factorization

Each of the factor pairs from the Collaboration can be used to rewrite 24 using more than one pair of factors.

For example, the first factor pair was $2 \cdot 12$, so 24 can be rewritten as $24 = 2 \cdot 12$.

Because 12 can be written as $2 \cdot 6$, 24 can also be rewritten as $24 = 2 \cdot 2 \cdot 6$.

1. Since 6 can be expressed as _____, then 24 can be rewritten as $24 = \underline{\hspace{2cm}}$.

This is called the **prime factorization** of 24 because all of the factors are **prime numbers**.



A **prime number** is a whole number greater than 1 that is not divisible by any whole number other than 1 and itself.

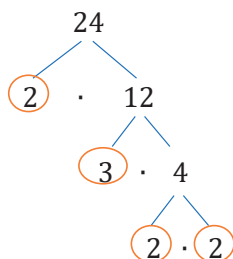
A **prime factor** is a factor that is a prime number.



Prime factorization is the expression of a number as the product of prime factors.

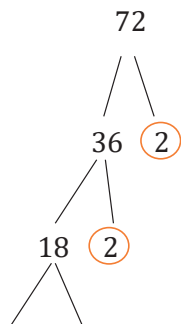
2. Cory determined the prime factorization of 24 using different factor pairs.

- a. Explain whether Cory's method changed the final result.



- b. Why did Cory circle the values he did?

3. Find the prime factorization of 72 by completing the work below.



The prime factorization of 72 is _____.

4. Consider the expression $2 \times 2 \times 2 \times 3 \times 3 \times 5$.

- a. What is the product of $2 \times 2 \times 2 \times 3 \times 3 \times 5$?

- b. The prime factorization of _____ is $2 \times 2 \times 2 \times 3 \times 3 \times 5$.

Numbers with multiple repeating prime factors can be written using exponents. For example, the prime factorization of 360 can be written as $2^3 \times 3^2 \times 5$.

5. Why is it helpful to write the prime factorization with exponents?



6.2.4 Guided Practice: Prime Factorization

- Find the prime factorization of each value. Use exponents where appropriate.
 - 132
 - 270
 - 882
- Sophie said the prime factorization of 210 is $3 \cdot 7 \cdot 10$.
 - Determine the error Sophie made. Then, write her a brief note explaining her error and how she can learn from it.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

- What is the prime factorization of 210?



6.2.5 Your Turn!

- Find the prime factorization of each value. Use exponents where appropriate.
 - 904
 - 1,125
 - 80
- Which number includes the largest exponent when written in prime factorization? Explain your answer.

5,670

64

- Find a number that meets the following conditions.
 - The value of the number is less than 150.
 - One of its prime factors has an exponent of 3.
 - When expressed in exponential form, its prime factorization has only two unique bases.



6.2.6 Wrap-Up: Ticket Out the Door

1. Find the prime factorization of each value. Use exponents where appropriate.

a. 168

b. 1,040

c. 144

Unit 6, Lesson 3: Adding Integers Using Integer Chips



Benchmark

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.



Learning Targets

- ☐ I can express integer addition using integer chips.
- ☐ I can add integers using integer chips.



6.3.1 Warm-Up: Below Sea Level

Dr. Ernest Everett Just was a pioneering African American biologist who conducted research at Woods Hole Marine Biological Laboratory in Massachusetts. He published two books and more than 70 scientific papers, most of which focused on the cell. Today, marine biologists use submarines to study marine life.



A submarine explorer begins at sea level and descends at a rate of 150 feet per minute.

1. Sketch a picture that shows how far below sea level the submarine will be in 3 minutes.
2. Add a bird that is flying 1,000 feet above sea level to your picture.
3. What number represents sea level?
4. Complete the statement.

The location of the submarine is represented by a

- ☐ positive value,
- ☐ negative value,

while the location of the bird is

represented by a

- ☐ positive value.
- ☐ negative value.



6.3.2 Guided Instruction: Zero Pairs

Positive and negative values are often useful in the contexts of money, banking, and accounting.

- In the context of money,

☐ positive
☐ negative

 numbers can be used to represent money spent or money owed to someone.
- ☐ Positive
☐ Negative

 numbers can be used to represent money earned.

Money can be used to learn how to add integers.

- A value of $+1$ means that you _____ \$1.
- A value of -1 means that you _____ \$1 or _____ \$1.

The table below shows a positive integer chip, which represents earning \$1, and a negative integer chip, which represents spending \$1.

Earn	Spend

Integer chips can also be used to represent integers.



Integers are whole numbers and their opposites.

A positive number can be written with a positive sign, $+1$ or $(+1)$, or it can be written without any sign, 1 .

A negative number can be written as -1 , (-1) or -1 .

- Complete the table. In the last column, sketch the integer chips that represent the phrase and the integer. Place a $+$ sign in a circle to represent $+$ and place a $-$ sign in a circle to represent $-$.

Phrase	Integer	Integer Chips
Earned \$6	$(+6)$	
Spent \$3	-3	

- Complete the table by sketching integer chips for each phrase.

Earned \$4	Spent \$4	Earned \$4, Then Spent \$4

- Joleen earned \$4, then spent \$4. How much money does she have left?

The last column in the table and problem above both represent a **zero pair**.

A **zero pair** is a pair of numbers whose sum is zero.

- Complete the table by drawing the negative integer chips that create a zero pair. In the **sum** column, draw all of the chips from both columns.

Positive	Negative	Sum
		

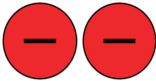
Each zero pair can be represented by an equation. The box shows two mathematical equations that represent the zero pair of $+$ and $-$.



Zero Pair as a Mathematical Equation

$$+1 + (-1) = 0 \quad \text{or} \quad 1 + -1 = 0$$

- Write an equation that represents the **sum** column in the previous table.
- Complete the table by drawing the positive integer chips that create a zero pair. In the **sum** column, draw all of the chips from both columns.

Positive	Negative	Sum
		

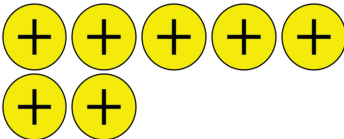
- Write an equation that represents the **sum** column in the previous table.





6.3.3 Exploration: Adding Integers with the Same Sign

1. Work with your partner to complete the table.

Mathematical Expression	Money Sentence	Integer Chips	Sum
	I earn \$5 babysitting and another \$2 for cleaning the dishes.		
$3 + 2$			
$-4 + -3$			
$(-2) + (-1)$			

2. With your partner, discuss the patterns you notice about the signs of the sums when adding integers with the same sign.
3. Were any zero pairs created when adding integers with the same sign? Explain.



6.3.4 Bring It Together: Adding Integers with the Same Sign

1. Complete the visual model of the rule of adding two positive integers by drawing the integer chip that represents the **sign** of the sum.

Visual Model of Addition	Sign of Sum
	

2. Complete the visual model of the rule of adding two negative integers by drawing the integer chip that represent the **sign** of the sum.

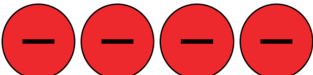
Visual Model of Addition	Sign of Sum
	



6.3.5 Exploration: Using Zero Pairs to Add Integers with Opposite Signs

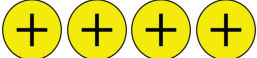
It is possible to add a positive number and a negative number. To find the answer using integer chips, combine a $+$ and $-$ to create a zero pair. Make as many zero pairs as possible. Then, determine the number of chips that are left over.

1. Complete the table to show how the sum $5 + (-4)$ can be represented using integer chips.

Step	Response
Represent the first addend with integer chips.	
Represent the second addend with integer chips.	
Determine the number of zero pairs created, if any.	
Cross out the integer chips that represent zero pairs to indicate they are eliminated from the model.	
Determine the sum using the number of integer chips remaining.	$5 + (-4) = \underline{\quad}$

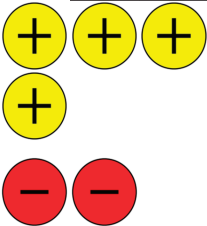
2. What do you notice about the remaining integer chips and the sign of the sum?

3. Complete the table independently to show how the sum $4 + (-7)$ can be represented using integer chips. Check your work with your partner.

Step	Response
Represent the first addend with integer chips.	
Represent the second addend with integer chips.	
Determine the number of zero pairs created, if any.	
Cross out the integer chips that represent zero pairs to indicate they are eliminated from the model.	
Determine the sum using the number of integer chips remaining.	$4 + (-7) = \underline{\quad}$

4. What do you notice about the remaining integer chips and the sign of the sum?

5. Work with your partner to complete the table. The first row is completed.

Mathematical Expression	Money Sentence	Integer Chips	Sign of Sum	Sum
$4 + (-2)$	I earn \$4 and spend \$2.		+	+2
$3 + (-6)$				
$-7 + 5$				
$(-1) + 8$				

6. Complete the table below independently. Check your work with your partner.

Mathematical Expression	Number of Zero Pairs	Sign of Sum	Sum
$(-10) + 12$	10	+	2
$12 + -7$			
$(-11) + 4$			
$8 + -8$			
$-10 + 3$			
$9 + (-2)$			

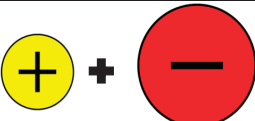
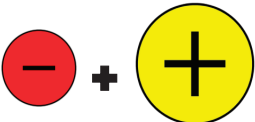
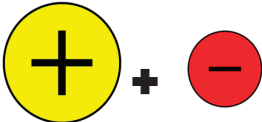
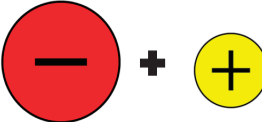
7. What do you notice about expressions where the sum is a positive number?
8. What do you notice about expressions where the sum is a negative number?



6.3.6 Bring It Together: Adding Integers with Opposite Signs

The table below shows visual models of rules for adding integers with opposite signs. In each visual model, the larger circle symbolizes the integer represented by the most integer chips.

1. Write an example of the rule. The first row has been completed.

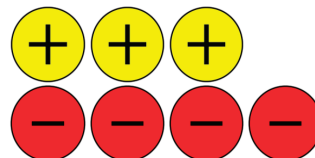
Visual Model of Addition	Sign of Sum	Example
		$2 + (-5) = -3$
		
		
		

2. Give two observations about the rules for adding integers with opposite signs.



6.3.7 Wrap-Up: Ticket Out the Door

1. Simone created the sketch below to represent the expression $4 + (-3)$.



Her sketch is incorrect. Explain why and create a new sketch to find the sum.

2. Complete the table for the expression below.

Expression	$3 + -5$
Integer Chips	
Money Sentence	
Sign of the Sum	
Sum	

Unit 6, Lesson 4: Adding Integers Using Number Lines



Benchmark

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.



Learning Target

- ☐ I can add integers using a number line.



6.4.1 Warm-Up: Speed Is Not Important

How fast you perform calculations in mathematics is not at all related to how “good” you are at math. Fast calculation is not important. What matters is deep thinking and understanding.

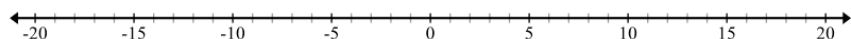
1. Describe a time when you felt rushed to perform calculations on a math assignment or assessment.
2. If you were given as much time as you needed in the situation you described, how would that have affected your feelings and performance on completing the assignment or assessment?



6.4.2 Refresher: Plotting Numbers on a Number Line

1. Plot each of the following integers on the number line below. Label each point with the letter indicated.

$$A = 4 \quad B = -4 \quad C = -16 \quad D = 8 \quad E = 18 \quad F = -19$$



2. Complete the statements.

- a. When moving from left to right ($\rightarrow$) on the number line, the values
☐ increase.
☐ decrease.
 This can be described as moving in a
☐ positive
☐ negative
 direction.
- b. When moving from right to left ($\leftarrow$) on the number line, the values
☐ increase.
☐ decrease.
 This can be described as moving in a
☐ positive
☐ negative
 direction.



6.4.3 Guided Instruction: Rules for Adding Integers on a Number Line

In addition to integer chips, number lines are another helpful tool you can use to model adding integers.

1. Complete the following to model integer addition.

$$-3 + (-2) = \underline{\hspace{2cm}}$$

Integer Chip Model	Number Line Model

$$4 + (-7) = \underline{\hspace{2cm}}$$

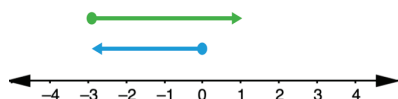
Integer Chip Model	Number Line Model

2. The sum of integers can be represented using rays on a number line. A ray has a single endpoint and extends in a specified direction based on the sign of the integer.



- A ray that represents a positive integer has its endpoint at 0, with the arrow extending to the _____.
- A ray that represents a negative integer has its endpoint at 0, with the arrow extending to the _____.
- To represent a sum, draw a ray for the first integer that starts at 0 and extends in the direction indicated by the _____ of the first integer. It ends at the location of the first integer on the number line.
- Then, draw a ray for the second integer that starts where the first ray's arrow ends and extends in the direction indicated by the _____ of the second integer. It ends after moving the number of units indicated by the second integer.
- The sum of the two integers is the integer where the second ray ends.

3. The diagram shown represents $-3 + 4 = +1$



Give an example or explain when a number line model would be more useful than using integer chips.



6.4.4 Collaboration: Commutative Property

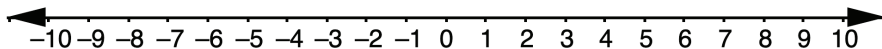
The **commutative property of addition** states that the order in which addition is performed does not change the value of the expression.

With your partner, add integers using the commutative property.

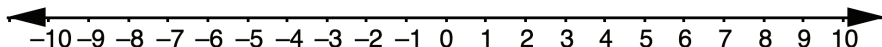
1. Complete the table by sketching the integer chip model for the first column, the number line model for the third column, and answering the question in the last column.

Integer Chip Model	Number Line Model	Commutative Property Number Line Model	Yes/No
$-3 + 4$	$-3 + 4$ 	$4 + (-3)$ 	Is the sum of $-3 + 4$ the same as the sum of $4 + (-3)$?
$-4 + 2$	$-4 + 2$ 	$2 + (-4)$ 	Is the sum of $-4 + 2$ the same as the sum of $2 + (-4)$?

2. Show the model for $(-7) + 9$ on the number line below.



3. Show the model for $9 + (-7)$ on the number line below.



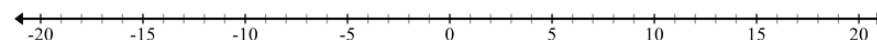
4. Discuss with your partner how the two number line models for $(-7) + 9$ and $9 + (-7)$ show that $(-7) + 9 = 9 + (-7)$. Write a summary of your reasoning.



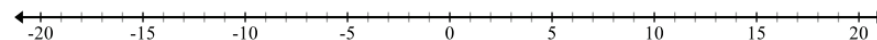
6.4.5 Try It: Adding Integers Using Number Lines

1. Find each sum. Use the number line to model the addition.

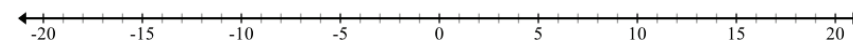
a. $8 + (-18) = \underline{\hspace{2cm}}$



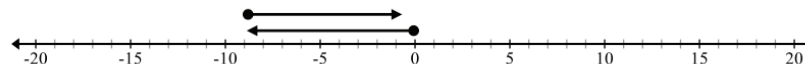
b. $-7 + (-5) = \underline{\hspace{2cm}}$



c. $-3 + 20 = \underline{\hspace{2cm}}$



2. Write the numerical expression for the model below. Then, find the sum.

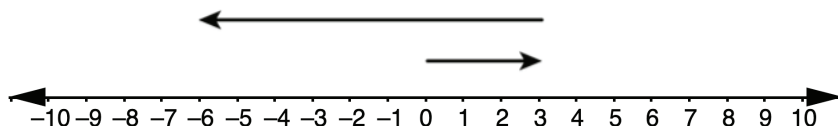




6.4.6 Your Turn!

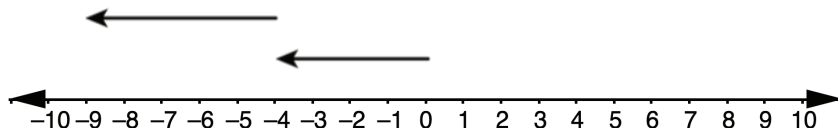
1. For each number line model, write the addition expression represented and find the sum.

a.



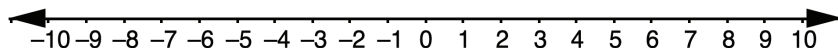
Addition Expression	Sum

b.



Addition Expression	Sum

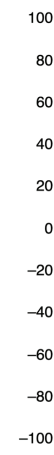
2. Use the number line to show the sum of $-8 + 5$. Then, complete the equation.



$$-8 + 5 = \underline{\hspace{2cm}}$$

3. In a single day, the temperature in degrees Fahrenheit ($^{\circ}\text{F}$) of a town in Canada began at a low of -20°F and increased 40°F by noon.

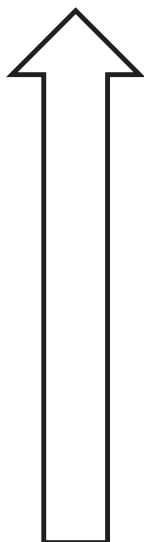
- Plot a point at the low temperature.
- Draw a ray along the number line to represent the increase of 40°F .
- Which direction does the arrow extend to show the increase in temperature?
- What was the temperature in the town by noon?





6.4.7 Wrap-Up: Reflection

1. Color in the arrow up to the statement that best describes your current level of confidence with using a number line to add integers.



I'm very confident! I can teach anyone how to add integers.

I can help others add integers.

I can add integers on my own.

I can add integers with assistance.

I'm having trouble adding integers.

I'm not sure what integers are.

Unit 6, Lesson 5: Adding Integers



Benchmark

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.



Learning Target

- ☐ I can add integers with procedural fluency.



6.5.1 Warm-Up: True Equations

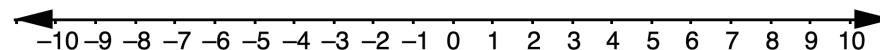
1. Using integers between -3 and 3 one time each, fill in the blanks to make the equation true.

$$\square + \square = \square$$



6.5.2 Exploration: Additive Inverses

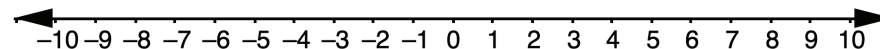
1. Use the number line to show the sum of $-4 + 4$. Then, complete the equation.



$$-4 + 4 = \underline{\hspace{2cm}}$$

2. Represent the sum of $-4 + 4$ using integer chips.

3. Use the number line to show the sum of $7 + (-7)$. Then, complete the equation.



$$7 + (-7) = \underline{\hspace{2cm}}$$

4. Represent the sum of $7 + (-7)$ using integer chips.



Existence of Additive Inverses

For every a there exists $(-a)$ so that $a + (-a) = 0$ and $(-a) + a = 0$.

Additive inverses are another way to represent the concept of a zero pair.

5. What number is the additive inverse of -15 ?

6. What number is the additive inverse of 27 ?

7. Choose any positive integer and write it in the blank: ____.

Use your number to fill in the blanks in the statement.

For the value ____, there exists $(- \text{____})$ so that $\text{____} + (- \text{____}) = 0$ and $(- \text{____}) + \text{____} = 0$.

8. Write your partner's positive integer: ____.

Use your partner's number to fill in the blanks in the statement.

For the value ____, there exists $(- \text{____})$ so that $\text{____} + (- \text{____}) = 0$ and $(- \text{____}) + \text{____} = 0$.



6.5.3 Guided Instruction: Adding Integers

Understanding the existence of additive inverses can be helpful when adding integers with larger absolute values.

1. Complete each statement based on your understanding of adding integers so far.

a. When adding two positive integers, the sum is

- ☐ positive.
- ☐ negative.

b. When adding two negative integers, the sum is

- ☐ positive.
- ☐ negative.

When adding a positive and a negative integer, the sum will be either positive or negative, depending on which value has the larger absolute value.

2. Consider the expression $40 + (-165)$.

a. What will be the sign of the sum?

- ☐ positive
- ☐ negative

b. Explain your reasoning.

When adding these integers, zero pairs are created.

- c. Use additive inverses to explain how many zero pairs are created.
- d. What remains after combining the additive inverses?

- e. Complete the equation.

$$40 + (-165) = \underline{\hspace{2cm}}$$

- 3. Complete the statements to describe how to find the sum of $-1,029 + (-38)$.

When adding $-1,029 + (-38)$, ☐ zero pairs
☐ no zero pairs are

created because the signs of the addends are

- ☐ the same.
☐ opposites.

The sum can be found by adding the

absolute values of the addends. Since both addends are

- ☐ positive,
☐ negative,

the ray for the second addend of the sum

will go farther to the ☐ right ($\rightarrow$)
☐ left ($\leftarrow$) of zero on a number line.

The sum of $-1,029 + (-38)$ is

- ☐ 991.
☐ -991.
☐ 1067.
☐ -1067.



6.5.4 Collaboration: Reasoning with Integers

- 1. Work with your partner to fill in the blanks with an integer to make each equation true.

$$\boxed{14} + \boxed{} = \boxed{-35}$$

$$\boxed{} + \boxed{-70} = \boxed{234}$$

$$\boxed{-136} + \boxed{} = \boxed{25}$$

- 2. Choose one equation and explain your thinking in determining the missing value.





6.5.5 Your Turn!

1. Find each sum.

a. $10 + (-204)$

b. $(-99) + (99)$

c. $904 + (-11)$

d. $-1 + 76$

e. $205 + (-643)$

f. $-151 + (-70)$

g. $-399 + (-405)$

h. $-80 + 606$

i. $703 + (-190)$

j. $-600 + (-78)$



6.5.6 Wrap-Up: Growth Mindset

1. Today, I am still unsure about _____

_____.

2. To fill in this gap, I intend to _____

_____.

Unit 6, Lesson 6: Subtracting Integers Using Integer Chips



Benchmark

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.



Learning Target

- ☐ I can use integer chips to subtract integers.



6.6.1 Warm-Up: When Will I Use This?

1. The speaker states, "It's not about what you learn, it's about what methods, tools, and tactics you have to develop in order to solve the problem that you may never see again for the rest of your life ..."
What does he mean by "methods, tools, and tactics"?



2. The quote continues to say, "... but you will see other problems where these methods and tools will become immensely valuable to you." Where might these methods, tools, and tactics help you in everyday life?





6.6.2 Exploration: Subtracting Integers

1. In the first column, a larger circle represents a number that has a larger absolute value than that of the smaller circle. Without discussing it with your partner, enter a numeric example for each visual model. Then, in the **"My Guess: Answer Will be + or -"** column, enter whether you think the result will be positive (+) or negative (-).

After completing all of your own examples and guesses, record your partner's examples and guesses.

Visual Model of Subtraction	My Example	My Guess: Answer Will be + or -	Partner's Example	Partner's Guess: Answer Will be + or -

2. Now, form a group of four by joining another pair of classmates. Discuss your guesses. Do all four of you agree on all of the problems? If not, list which problems you disagree on.
3. Tell your teacher when you are ready to test out the guesses with a calculator. Use the calculator to find the answer for one problem from each visual model. When each of you are in agreement with the rule, check the box in the column that represents the answer.

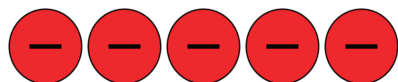
Visual Model of Subtraction				



6.6.3 Guided Instruction: Using Integer Chips to Show Subtraction

Integer chips can also be used to model subtraction.

To find the difference of $(-5) - 2$ using integer chips, begin by first arranging 5 negative integer chips on your desk, as shown below.



1. Draw a picture of how you think integer chips can be used to represent subtracting positive 2 from the 5 negative integer chips that are already on your desk.

Consider using zero pairs. It is okay if you do not know; just try your best.



Additive Identity Property of Zero

$$a + 0 = a \quad \text{or} \quad 0 + a = a$$

When adding 0 to a value, the value of the expression does not change. In a model, adding zero pairs represents adding 0. The value of the model does not change, but the integer chips that are available do.



2. Complete the table to show how $(-3) - 6$ can be represented using integer chips.

Begin by representing the first integer, -3 , with integer chips.	
Determine what should be subtracted from the model.	6 ☐ positive ☐ negative integer chips need to be subtracted from the model
If the chips that need to be removed from the model are not available, add the number of zero pairs needed to be able to do so.	
Cross out the integer chips from the model drawn above to represent the subtraction.	
Find the value of the expression based on the integer chips remaining.	$-3 - 6 = \underline{\hspace{2cm}}$

3. Look at your guess for representing $(-5) - 2$. Using the directions in the table for $(-3) - 6$, determine if your guess was correct. If your guess was correct, explain how you decided you should add two zero pairs to the model. If your guess was not correct, explain what confused you about the zero pairs.

4. Complete the table to show how $4 - 6$ can be represented using integer chips.

Begin by representing the first integer with counters.	
Determine what should be subtracted from the model.	— ☐ positive ☐ negative integer chips need to be subtracted from the model
If the chips that need to be removed from the model are not available, add the number of zero pairs needed to be able to do so.	
Cross out the integer chips from the model drawn above to represent the subtraction.	
Find the value of the expression based on the integer chips remaining.	$4 - 6 = \underline{\hspace{2cm}}$



6.6.4 Guided Practice: Subtraction with Integer Chips

1. Complete the table to show how $(-2) - (-5)$ can be represented using integer chips.

Begin by representing the first integer with integer chips.	
Determine what should be subtracted from the model.	— ☐ positive ☐ negative integer chips need to be subtracted from the model
If the chips that need to be removed from the model are not available, add the number of zero pairs needed to be able to do so.	
Cross out the integer chips from the model drawn above to represent the subtraction.	
Find the value of the expression based on the integer chips remaining.	$-2 - (-5) = \underline{\hspace{2cm}}$

2. Use integer chips to represent how $-4 - -4 = 0$.



3. Jennifer borrowed money from Amber twice. The first time, it was \$1 for a bottle of water. The second time, it was \$5 for ice cream. Jennifer owes Amber a total of \$6. Later, Amber told Jennifer not to worry about paying her back \$2 of the \$6. The amount that Jennifer still owes Amber can be represented by the expression $-6 - (-2)$.

- a. Draw integer chips to represent $-6 - (-2)$ and the value of the expression.
- b. Determine the amount that Jennifer still owes Amber.



6.6.5 Your Turn!

1. Use integer chips to find the value of each expression.

a. $-11 - 6$

b. $-4 - (-7)$

c. $5 - (-8)$

d. $9 - (-3)$

e. $-1 - 2$

f. $-12 - 4$



6.6.6 Wrap-Up: Cool Down

1. Choose two of the visual models of subtraction that you are sure will result in a positive answer.

- a. Draw an asterisk (*) next to both in the table.

- b. Give an example of them below.

2. Choose one visual model that is a challenge for you.

- a. Draw an exclamation point (!) next to it in the table.

- b. Explain using an example why it is a challenge.

Visual Model of Subtraction

Unit 6, Lesson 7: Subtracting Integers Using Number Lines



Benchmark

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.



Learning Targets

- ☐ I can subtract integers using a number line.
- ☐ I can use number lines to understand how the rules for adding integers are related to the rules for subtracting integers.



6.7.1 Warm-Up: Bell Work

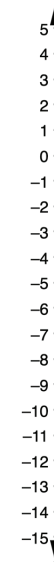
1. Draw integer chips that represent the expression below and evaluate.

$$7 - (-3)$$



6.7.2 Guided Instruction: Subtracting Integers Using a Number Line

1. The temperature in degrees Fahrenheit ($^{\circ}\text{F}$) of a city in Alaska was -4°F . After sunset, the temperature dropped 6°F .
 - a. Represent the change in temperature on the number line.
 - b. Write the subtraction number sentence that is represented from the context and on the number line.
 - c. After the decrease in temperature, what was the temperature?
 - d. Represent the temperature change using integer chips to verify the solution.



2. Complete the statements to explain how to represent subtraction on a horizontal number line.

a. To find the difference of $-9 - (-30)$, start by plotting

- ☐ 9
- ☐ -9
- ☐ 30
- ☐ -30

on the number line first.

b. The subtraction sign indicates moving to the

- ☐ right
- ☐ left

on the number line, but it is followed by

a negative sign with 30, which requires moving in the opposite direction.

c. Therefore, the answer will be found by moving

- ☐ 9
- ☐ 30

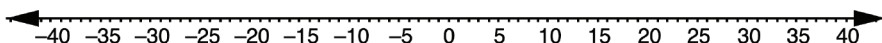
units to the

- ☐ right
- ☐ left

of -9 on the number

line.

3. Show the subtraction on the number line below.



The difference of $-9 - (-30)$ is _____.



6.7.3 Collaboration: Relating Integer Addition and Subtraction

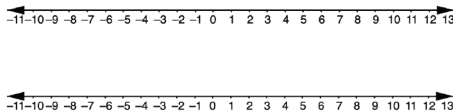
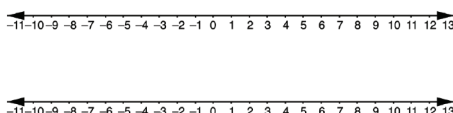
Three contexts are given below. Choose one context to work on with your partner. Circle your choice.

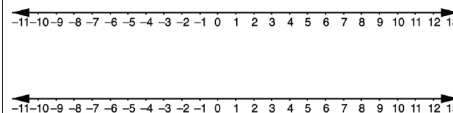
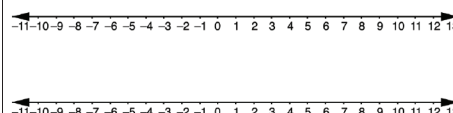
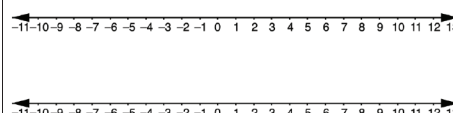
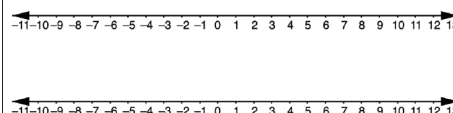
Context A	Tamara didn't realize she had a balance of \$19 in her bank account when she used her bank card to purchase a \$50 purse. When the purchase is processed by the bank, what will be her new account balance?
Context B	The elevation of the top floor of a parking garage in Death Valley, California, is 282 feet (ft) below sea level. A car that leaves the parking garage from the top floor loses 72 ft in elevation before reaching the street. What is the elevation of the street?
Context C	Ginny mows lawns for her summer job. Before she started mowing, she spent \$32 on gas for the lawn mower. Then, she spent another \$17 on other supplies she bought at the hardware store. What were Ginny's total expenses?

For the context selected, represent the mathematics in the context in each of the following ways.

1. Draw a number line to represent the given context.

- Write two equations that can be used to find the solution to the question in the context. One equation should use subtraction and the other should use addition.
- Create a group of four by joining with another pair of students. Verify your work by exchanging with the other pair of students.
- Continue working with the group of four to complete the table.

Example Subtraction Problem	Example Rewritten as an Addition Problem	Number Lines: Show the subtraction on the top number line and the addition on the bottom number line.
$10 - 7 = 3$	$10 + (-7) = 3$	
$7 - 10 = -3$		

$8 - -3 = 11$		
$-5 - 3 = -8$		
$-7 - -15 = 8$		
$-9 - -7 = -2$		



When subtracting integers, problems can be rewritten as an addition problem using the additive inverse of the second integer.

- Describe one advantage of changing a subtraction problem to an addition problem.
- Complete the table by rewriting the second integer in the subtraction model as its additive inverse and indicating the sign of the value of either expression. The first row has been completed. Once your group is done, split back into pairs and check your table with a different pair of students.

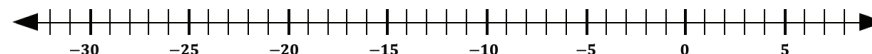
Visual Model of Subtraction	Visual Model of Addition	Sign of the Answer
		+



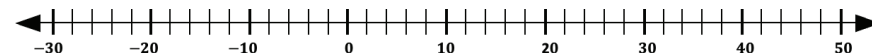
6.7.4 Your Turn!

- Represent each subtraction statement on a number line and find the value of the expression.

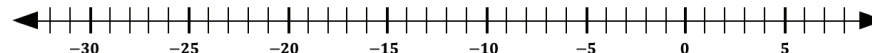
a. $-19 - 11 = \underline{\hspace{2cm}}$



b. $12 - (-28) = \underline{\hspace{2cm}}$



c. $(-14) - (-15) = \underline{\hspace{2cm}}$





6.7.5 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.

1. I can subtract integers using a number line.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

2. I can use number lines to understand how the rules for adding integers are related to the rules for subtracting integers.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

Unit 6, Lesson 8: Integer Subtraction



Benchmark

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.



Learning Target

- ☐ I can subtract integers with values greater than 100.



6.8.1 Warm-Up: Self-Reflection

For each statement, rate yourself on a scale of 1-5. A rating of 5 means “absolutely yes” and 1 means “absolutely not.”

Statement	Rating
I can subtract integers using integer chips.	
I understand how to show subtraction of integers on the number line.	
I can subtract integers without using integer chips or a number line.	
I can explain to someone else how to subtract integers.	



6.8.2 Guided Practice: Subtracting Integers

- Two expressions are shown.

Expression A	Expression B
$-191 + 77$	$-191 - (-77)$

With your partner, determine if the expressions are equivalent. Summarize your determination below.

- Annalise is on a trivia game show. Her score after the first round is -500 points. She misses the first question in the second round, losing an additional 125 points.

Kiersten said she could represent Annalise’s score using subtraction, while Dylan said it could only be represented with addition.

- Complete the table with the expression each student was thinking of.

Kiersten’s Subtraction Expression	Dylan’s Addition Expression
_____ - _____	_____ + _____

- Kiersten said Annalise’s score was -375 after missing the first question in the second round, but Dylan said it was -625 . Explain which student is correct.

3. Rewrite each subtraction equation as an equivalent addition equation.

Subtraction Equation	$513 - 382 = 131$	$957 - ^{-}136 = 1,093$
Addition Equation		

Addition and subtraction are **inverse operations**, which means they are operations that have the opposite effect of each other.

Understanding this relationship and the existence of additive inverses can be helpful when subtracting integers.



Existence of Additive Inverses

For every a there exists $(-a)$ so that $a + (-a) = 0$ and $(-a) + a = 0$.

4. Complete the statements below using this information.

A subtraction expression can be rewritten as an equivalent addition expression by changing the operation from subtraction to _____ and changing the second value to its _____.



6.8.3 Try It: Subtracting Integers

1. The table shows the beginning and ending elevations, in feet (ft), of six different hiking trails. The change in elevation along the trail can be found by subtracting the initial elevation from the final elevation.

$$(\text{final elevation}) - (\text{initial elevation}) = (\text{change in elevation})$$

- a. Write an expression that represents the change in elevation along each hiking trail. Then, write the value that represents the change in elevation.

The first row has been completed as an example.

Initial Elevation (ft)	Final Elevation (ft)	Difference Between Final and Beginning	Change in Elevation (ft)
400	900	$900 - 400$	500
115	68		
294	-121		
-200	610		
-163	-57		
-17	-87		

- b. Explain what a negative change in elevation means in this context.

2. Find each difference.

a. $-664 - 231$

b. $109 - (-115)$



6.8.4 Your Turn!

1. Evaluate each expression.

a. $740 - (-171)$

b. $569 - 143$

c. $-738 - (-212)$

d. $333 - (-204)$

e. $-877 - 814$

f. $-691 - 293$

g. $157 - (-120)$

h. $341 - (-504)$



6.8.5 Wrap-Up: Ticket Out the Door

1. Evaluate each expression.

a. $447 - (-702)$

b. $-834 - (-610)$

c. $-309 - 525$

Unit 6, Lesson 9: Multiplying Integers Using Models



Benchmark

MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.



Learning Target

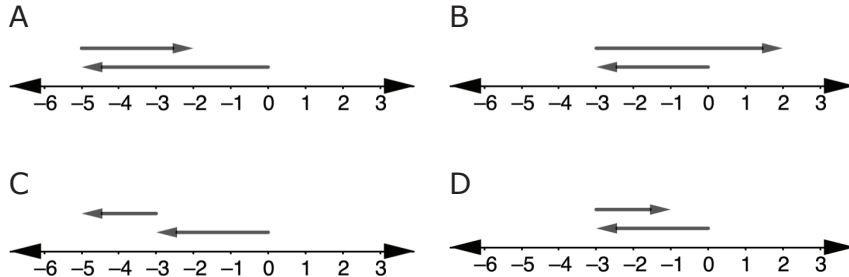
- ☐ I can multiply integers using integer chips, number lines, and patterns.





6.9.1 Warm-Up: Which One Doesn't Belong?

1. Four number lines modeling addition are shown.



Explain which one doesn't belong.



6.9.2 Exploration: Using Integer Chips to Represent Multiplication

1. The addition expression $7 + 7 + 7$ can be written as ____ $\cdot 7$.

This is because repeated addition of integers can be expressed as

- ☐ subtraction.
- ☐ multiplication.
- ☐ division.

2. The expression 5×2 can be interpreted as adding ____ groups of ____ or adding ____ groups of ____.

3. Use integer chips to represent one of these interpretations.

Recall that $\ominus = -1$ and $\oplus = +1$.

Draw a picture of your representation.

4. Describe how the integer chips represent the value of the expression 5×2 .

5. The expression $5 \times (-2)$ can be interpreted as adding _____ groups of _____ or adding _____ groups of _____.
6. Explain which interpretation of $5 \times (-2)$ you think would be easier to represent using integer chips.
7. Discuss your choice with your partner. Then, work together to use integer chips to represent one of the interpretations. Draw a picture of the representation.
8. Describe how the integer chips represent the value of the expression $5 \times (-2)$.
9. What is the value of the expression $5 \times (-2)$?



Commutative Property of Multiplication

$$a \times b = b \times a$$

The **commutative property of multiplication** states that the order in which multiplication is performed does not change the value of the expression.

10. Using the commutative property of multiplication, what is the value of the expression -2×5 ?



6.9.3 Exploration: Multiplying a Negative Integer by a Negative Integer

1. Complete the statement.

The expression, $(-5)(-2)$, indicates that _____ groups of _____ should be **removed**.

2. Use the steps below to discover how to represent $(-5)(-2)$ using integer chips.

Step 1: First, use your integer chips to create the set of zero pairs that will be needed to find $(-5)(-2)$. Draw the groups of zero pairs in the box below.



Step 2: Remove 5 groups of 2 negative integer chips.

Step 3: Remove any remaining zero pairs, if needed.

3. Complete the statement.

The value of expression $(-5)(-2)$ is _____.

4. With your partner, discuss whether or not you think the product when multiplying two negative values will always be positive. Explain your thinking.

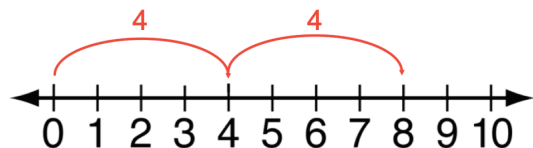




6.9.4 Guided Instruction: Multiplying Integers Using the Number Line

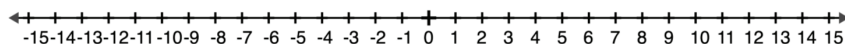
Recall that $(2)(4)$ means 2 groups with 4 units in each group. The product is the total number of objects.

This can be represented on the number line as shown.



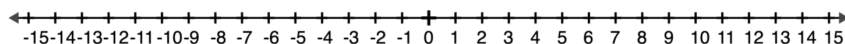
1. Represent each multiplication expression on the number line. Then, use it to complete the equation.

$$(2)(-4)$$



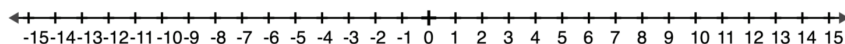
$$(2)(-4) = \underline{\hspace{2cm}}$$

$$(-2)(4)$$



$$(-2)(4) = \underline{\hspace{2cm}}$$

$$(-2)(-4)$$



$$(-2)(-4) = \underline{\hspace{2cm}}$$



6.9.5 Guided Practice: Multiplying Integers Using Models

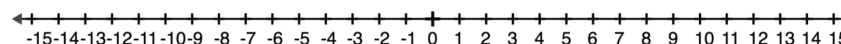
1. Use integer chips to model and evaluate each expression.

$$\text{a. } (3)(-5) =$$

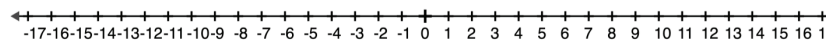
$$\text{b. } -2 \times -2 =$$

2. Use a number line model to evaluate each expression.

$$\text{a. } (8)(-1) =$$



$$\text{b. } (-4)(-4) =$$



3. Which model for multiplication of integers do you prefer: multiplication chips or the number line?



6.9.6 Your Turn!

1. Find the value of each expression using your preferred method.

a. 2×-8

b. $-6 \cdot (-3)$

c. $(-7)(-6)$

d. $5(-3)$

e. -9×-4

f. $7 \cdot 5$



6.9.7 Wrap-Up: Cool Down

1. Complete each rule by drawing the sign of the product as an integer chip.

Visual Model of Multiplication	Sign of Product
 $\times$ 	
 $\times$ 	
 $\times$ 	
 $\times$ 	

Unit 6, Lesson 10: Dividing Integers Using Integer Chips



Benchmark

MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.



Learning Target

- ☐ I can use integer chips and patterns to discover rules for dividing integers.



6.10.1 Warm-Up: Notice and Wonder



1. Write down one thing you notice about the picture.
2. Write down one thing you wonder about the picture.
3. Isaac described the image as a math fail. Why do you think he described it that way?



6.10.2 Guided Instruction: Using Integer Chips to Represent Division

Recall that division is repeated subtraction.

For example, $24 \div 8$ can be expressed as $24 - 8 - 8 - 8$ because 3 groups of 8 can be subtracted from 24.

This shows that 24 is made up of ____ groups of 8.

The expression $24 \div 8$ can also be interpreted as dividing 24 into 8 equal-sized groups. In the representation shown, 24 positive integer chips are divided into 8 equal-sized groups.

+	+	+
+	+	+

+	+	+
+	+	+

+	+	+
+	+	+

+	+	+
+	+	+

The quotient is the result of the division, or the value of the integer chips in each group.

$$24 \div 8 = \underline{\hspace{2cm}}$$

1. Complete the table to describe the division, then draw the visual model using integer chips, and determine the quotient.

Division Expression	Written Description	Visual Model of Division	Quotient
$\frac{15}{5}$	Divide ____ into ____ equal-sized groups		
$-6 \div 2$	Divide ____ into ____ equal-sized groups		

When modeling division with integer chips, the divisor must be positive, as this indicates the number of groups needed. When the divisor is negative, the expression can be rewritten using opposites.

For example, $\frac{8}{-2}$ can be rewritten as $\frac{-8}{2}$ because $\frac{8}{-2} = \frac{-8}{2}$.

2.

Division Expression	Equivalent Division Expression	Written Description	Visual Model of Division	Quotient
$\frac{9}{-3}$		Divide ____ into ____ equal-sized groups		
$-12 \div -2$		Divide ____ into ____ equal-sized groups		

3. Use the pattern of the division in the tables to complete the rules by drawing the sign of the quotient as an integer chip.

Visual Model of Division	Sign of Quotient
$\frac{+}{+}$	
$\frac{+}{-}$	
$\frac{-}{+}$	
$\frac{-}{-}$	
$\frac{+}{+}$	
$\frac{-}{-}$	
$\frac{-}{-}$	



6.10.3 Collaboration: How Are Integer Multiplication and Division Related?

Jeannie and Solimar were discussing the relationship between $5 \times -2 = -10$ and $(-10) \div (-2) = 5$.

Solimar said, "The multiplication equation can be interpreted as 5 groups of -2 and the division equation can be interpreted as how many groups of -2 are in -10 . Isn't that basically the same thing?"

Jeannie replied, "I agree. I also think it is interesting to see how the rules seem to be related. In the multiplication equation, we are multiplying a positive by a negative and the answer is a negative. In the division equation, we do the reverse of the multiplication problem and find that a negative divided by a negative is a positive."

Solimar said, "Yeah, that's a cool connection. Do you think the rules are the same for multiplication and division?"

Jeannie said, "I think they might be related to each other."

1. Discuss Solimar and Jeannie's conversation with your partner.
2. Create two integer division equations that can be rewritten with the related multiplication equation.

Division Equation	Related Multiplication Equation



6.10.4 Bring It Together: How Are Integer Multiplication and Division Related?

- Write the missing factor in each given equation. Then, write the related division or multiplication equation.

Given Equation	Can be Thought of as...	Related Equation
$3 \times \underline{\quad} = 21$	$\longleftrightarrow$	$21 \div 3 = \underline{\quad}$
$\frac{-48}{4} = \underline{\quad}$	$\longleftrightarrow$	$(\underline{\quad})(4) = -48$
$-8 \times \underline{\quad} = 56$	$\longleftrightarrow$	$56 \div (-8) = \underline{\quad}$
$20 \div \underline{\quad} = -5$	$\longleftrightarrow$	$(-5)(\underline{\quad}) = 20$
$-\left(\frac{36}{4}\right) = \underline{\quad}$	$\longleftrightarrow$	$\underline{\quad} \times 4 = -36$

- How are rules for multiplication of integers related to the rules for division of integers?



6.10.5 Collaboration: Which One Doesn't Belong?

- Circle the expression that does not belong with the other three.

$$\frac{10}{-5}$$

$$\frac{-10}{5}$$

$$\frac{-10}{-5}$$

$$-\frac{10}{5}$$

- Explain why the expression does not belong.

- Verify your answers with your partner and, if necessary, resolve any differences.

Quotients can have negative signs in different places.

- The values $\frac{12}{-2}$, $\frac{-12}{2}$, and $-\left(\frac{12}{2}\right)$ are equivalent. Explain why they are equivalent using your understanding of dividing integers.

- Discuss your explanation with your partner. If necessary, edit your explanation based on your discussion.

- Create two equivalent expressions for $21 \div (-7)$.

- $\frac{-40}{-8}$ is not equivalent to $-\left(\frac{40}{8}\right)$. Explain why not.



6.10.6 Guided Practice: Dividing Integers

1. Find each quotient.

a. $12 \div -4 = \underline{\hspace{2cm}}$

b. $\frac{-63}{-9} = \underline{\hspace{2cm}}$

c. $-\left(\frac{28}{4}\right) = \underline{\hspace{2cm}}$

d. $60 \div 5 = \underline{\hspace{2cm}}$

e. $-\left(\frac{-30}{-5}\right) = \underline{\hspace{2cm}}$

2. Create two expressions equivalent to $\frac{-20}{10}$.

3. What is the value of the expressions above?



6.10.7 Your Turn!

1. In each column below, create three different division expressions that would result in the given quotient.

-4	10	-7
1.	1.	1.
2.	2.	2.
3.	3.	3.



6.10.8 Wrap-Up: Reflection

1. Use the emoji scale to indicate your level of confidence with understanding and using the rules for multiplying and dividing integers.



0 1 2 3 4 5 6 7 8 9 10



Unit 6, Lesson 11: Multiplying and Dividing Integers



Benchmark

MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.



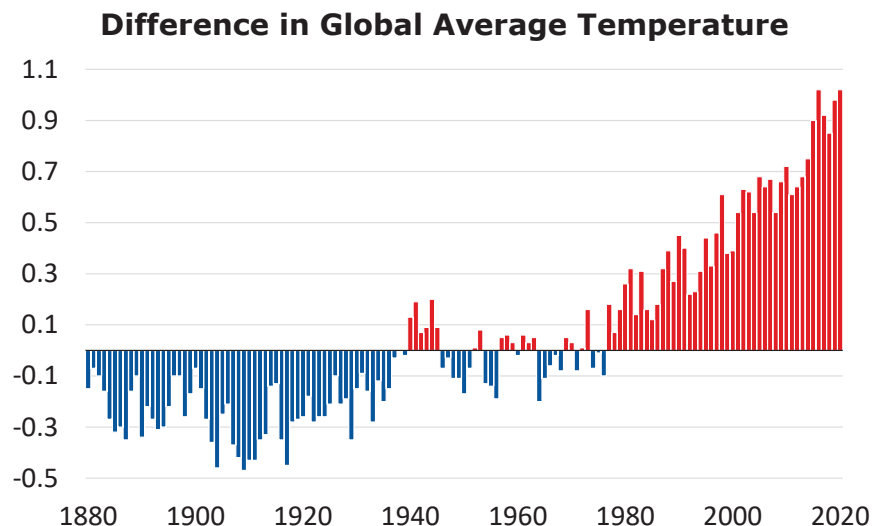
Learning Target

- ☐ I can multiply and divide integers with procedural fluency.



6.11.1 Warm-Up: What's Going on in This Graph?

The graph shows the average global temperatures between 1880 and 2020 compared to the middle of the 20th century. All temperatures are in degrees Celsius ($^{\circ}\text{C}$).



1. What are you curious about after examining the graph?
2. Imagine this graph is included in a newspaper article. Write a catchy headline that captures what's going on in the graph.



6.11.2 Refresher: Multiplying and Dividing Integers

1. Work with your partner to complete the tables. The last row of each table is missing both the rule and an example equation.

Visual Model of Multiplication	Sign of Product	Example Equation
$\times$		
$\times$		
		$-14 \cdot 4 = -56$

Visual Model of Division	Sign of Quotient	Example Equation
$\div$		
$\div$		
		$\frac{24}{4} = 6$



6.11.3 Collaboration: Multiplying and Dividing Integers

1. The table shows two different products. Work with your partner to find five different factor pairs that, when multiplied, will result in the product. The factors cannot include 1 or -1 .

Product	Factor	Factor
(-90)		
42		

2. Explain a strategy or an approach that each partner suggested.
3. Explain which of the products, (-90) or 42 , was the easiest to find factor pairs for and why.

4. The table shows two different quotients. Work with your partner to find four different pairs of dividends and divisors that result in the given quotient when they are divided. The numbers 1 or -1 cannot be used.

$$\text{dividend} \div \text{divisor} = \text{quotient}$$

Dividend	Divisor	Quotient
		(-8)
		13

5. Explain a strategy or an approach that each partner suggested.
6. Explain which of the quotients, (-8) or 13 , was the easiest to find dividends and divisors for and why.





6.11.4 Guided Practice: Finding Missing Factors, Products, or Quotients

1. Fill in the missing values to make each equation true.

a. $(-7) \cdot \underline{\hspace{1cm}} = -14$

b. $\underline{\hspace{1cm}} \div (-5) = 7$

c. $\frac{-40}{8} = \underline{\hspace{1cm}}$

d. $\underline{\hspace{1cm}} \times (-9) = 63$

2. Consider the equation shown. Phillip said -12 is the missing factor while Alicia claimed it is 12 .

$$(-1) \cdot (-2) \cdot \underline{\hspace{1cm}} = 24$$

Explain which student is correct using rules for multiplying integers.



6.11.5 Your Turn!

1. Find each product or quotient.

-11×9	$-4 \cdot (-12)$	$-100 \div 25$	$\frac{-60}{-5}$
$96 \div (-6)$	$(-5) \times (-13)$	$\frac{84}{-12}$	$(-7)(-7)$
$110 \div (-11)$	$(-6)(-24)$	$9 \times (-15)$	-18×96
$26 \div (-13)$	$150 \div (-3)$	$-39 \times (-4)$	$\frac{-88}{-4}$



6.11.6 Wrap-Up: Reflection

1. Complete the learning period based on today's lesson.

One question I would like to have answered ...

One mistake I learned from today ...

One thing I am not sure about ... YET!

One thing I am confident I know is ...



Unit 6, Lesson 12: Multiplying and Dividing Integers with Procedural Fluency



Benchmark

MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.



Learning Target

- ☐ I can multiply and divide integers with procedural fluency.



6.12.1 Warm-Up: Thinking Algebraically

A number line is shown. The numbers 0 and 1 are marked on the line, along with two other numbers, a and b .



1. Circle one expression from those shown below that will have a negative value.

$a - 1$	$a - 2$	$a + b$	$a - b$	$ab + 1$
---------	---------	---------	---------	----------

2. Explain how you know the expression selected will have a negative value based on the number line.

3. Find a classmate who chose a different expression that will have a negative value.

Record their expression and explanation for why they chose it in the table. Then, have them initial that you recorded it correctly. *You are the only person who should write in the first two columns of the table.*

Classmate's Selected Expression	Classmate's Explanation for Why the Value is Negative	Initials



6.12.2 Collaboration: Multiplying Multiple Integers

1. Work with your partner to find the product of each expression.

a. $5(-9)$	b. $-6(-4)$	c. $0(-3)$
d. $(-1)(-1)(3)$	e. $4 \cdot (-2) \cdot (5)$	f. $-2(-2)(-2)(-2)$
g. $(-1) \times (-2) \times 3 \times (-2) \times (-1) \times 2$		

2. In expressions d-g, more than two factors are being multiplied.

Discuss with your partner how the rules you developed previously for multiplying two integers could be extended for multiplying more than two integers.

Write at least one rule that would apply in this situation.



6.12.3 Activity: Integer War – Multiplication Edition

Set Up:

- Remove all jokers and face cards (Kings, Queens, and Jacks) from the deck.
- With your partner, determine who is the dealer. The dealer deals the entire deck of cards to each player, face down.

Rules:

- The black cards (Spades ♠ and Clubs ♣) represent positive values.
- The red cards (Diamonds ♦ and Hearts ♥) represent negative values.
- Aces represent either 1 or -1 , while the other cards represent the positive or negative value indicated on the card.

How to Play:

- To begin, each player will flip over one card from their deck and place it in the center.
- The first player who figures out the product of the two integers keeps both cards and adds them to the bottom of their stack.
 - NOTE: If a player says an incorrect product, the other player responds for a chance to win the hand before the first player can change their answer.
- Repeat this process until time is called or until one player has all of the cards in the deck.



6.12.4 Guided Practice: Finding Products or Quotients

1. Find each product or quotient.

a. $10 \times (-65)$

b. -19×99

c. $-812 \div (-7)$

d. $\frac{208}{-16}$

e. $-125 \div (-5)$

f. $(-21)(-16)$



6.12.5 Your Turn!

1. Find each product or quotient.

a. -17×29

b. $41 \cdot (-19)$

c. $-184 \div 23$

d. $\frac{-175}{-5}$

e. $96 \div (-6)$

f. $(-51) \times (-31)$

g. $-256 \div (-16)$

h. $(-17) \cdot 704 \cdot (-1)$



6.12.6 Wrap-Up: Ticket Out the Door

1. Complete the table by writing a value in each blank that completes each related equation.

Multiplication Equation	...related to...	Division Equation
$(-12)(\underline{\quad}) = 504$	$\longleftrightarrow$	$504 \div \underline{\quad} = -12$
$\underline{\quad} \times 18 = -126$	$\longleftrightarrow$	$-126 \div 18 = \underline{\quad}$
$-50 \cdot \underline{\quad} = -650$	$\longleftrightarrow$	$-650 \div \underline{\quad} = -50$

Unit 6, Lesson 13: Evaluating Exponents



Benchmark

MA.6.NSO.3.3 Evaluate positive rational numbers and integers with natural number exponents.



Learning Targets

- ☐ I can evaluate integers with natural number exponents.
- ☐ I can evaluate a positive rational number raised to a power.



6.13.1 Warm-Up: Consider the YET

1. Consider the belief that “you can get better at whatever you want to get better at.” Explain whether you agree or disagree.
2. Consider the quote, “I’ve never heard of someone who took up running and didn’t get faster the more they did it.” That person just may not have been a fast runner ... YET.

Complete the statement to describe something you want to get better at.

I _____
_____...YET!





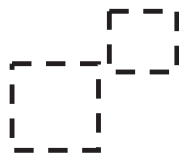
6.13.2 Exploration: Building on the Understanding of Exponents

Recall the vocabulary of exponential expressions.

exponent (or power) value
(of the
expression)

$$\underset{\text{base}}{3}^{\text{exponent (or power)}} = \underbrace{3 \cdot 3 \cdot 3 \cdot 3}_{\text{factors}} = 81$$

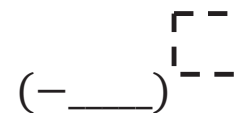
1. Think of a number that will result in a value of 16 when it is raised to a power less than 6.
2. Write the base and its exponent.



3. Compare your expression with your partner and check for accuracy.
4. If your partner found a different base and exponent that meets the description, write it below. If your partner wrote the same base and exponent as you, work together to find a second exponential expression that also works.



5. With your partner, choose one of the exponential expressions used and rewrite it, making the base negative.



6. Explain whether the negative base changes the value of the expression.
7. Discuss with your partner how you could change the exponent to result in a negative answer. Write a brief summary of your discussion below.



6.13.3 Bring It Together: Building on the Understanding of Exponents

When evaluating exponents with a negative base, the same rules of multiplication with integers apply.

1. In the expression $(-4)^2$, the base is _____ and the exponent is _____, resulting in the repeated multiplication sentence of _____.
2. Why do the results of $(-4)^2$ result in a positive number?
3. In the expression $(-4)^3$, the base is _____ and the exponent is _____, resulting in the repeated multiplication sentence of _____.
4. Why do the results of $(-4)^3$ result in a negative number?

5. For each expression, determine whether its value is positive or negative.

Expression	Positive or Negative Value?
$(-2)^2$	
$(-2)^3$	
$(-2)^4$	
$(-2)^5$	
$(-2)^{19}$	

6. When evaluating exponents with a negative base, the value of the expression will be

☐ positive
☐ negative

 if the exponent is an

☐ odd
☐ even

 number, and the value of the expression will be

☐ positive
☐ negative

 if the exponent is an

☐ odd
☐ even

 number.



6.13.4 Your Turn!

1. Evaluate each expression.

a. $(-3)^4$

b. $\left(\frac{15}{17}\right)^2$

c. $(-1)^5$

2. Rewrite each repeated multiplication sentence as an exponential expression. Then, find the value of the expression.

a. $(-12)(-12)(-12)$

b. 3.4×3.4

3. Determine if each equation in the table is true or false. Show your work to support your answer.

Equation	True or False?	Work
$16^1 = (-8)^2$		
$(-5)^3 = -5 \cdot 5 \cdot 5$		



6.13.5 Wrap-Up: Reflection

1. In this unit, you have learned multiple concepts about operations with integers. Rate yourself on the following skills based on your level of understanding.

Statement	Rating
	1 – I struggle with this. 2 – I have some skills. 3 – I am almost there. 4 – I have got this!
I can evaluate positive rational numbers and integers raised to a power.	
I can express composite whole numbers as a product of prime factors.	
I can add integers.	
I can subtract integers.	
I can multiply integers.	
I can divide integers.	